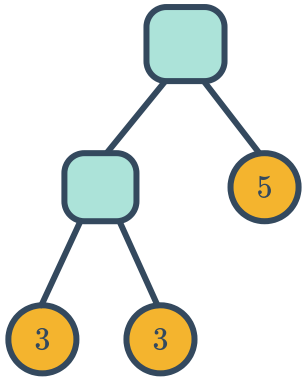
Factor trees

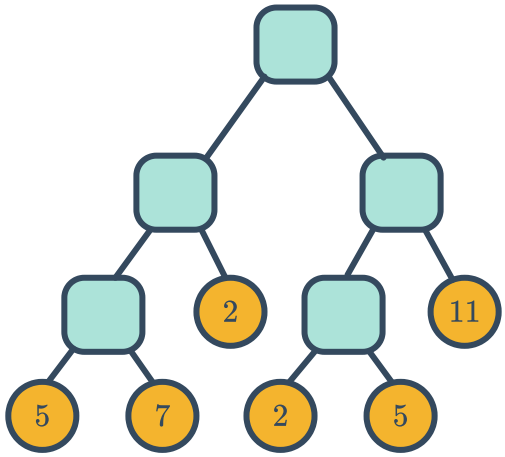
1 Select all the prime numbers from each of these lists:

a $3, 10, 12, 4, 5, 6$ **b** $14, 7, 15, 16, 13, 11$ **c** $21, 19, 27, 17, 22, 1$

2 A number has the following factor tree:
What is this number at the top of the tree?



3 A number has the following factor tree:
What is this number at the top of the tree?



4 Construct a factor tree for the following numbers:

a 6 **b** 12 **c** 28 **d** 42

e 81 **f** 144 **g** 200 **h** 360

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Prime factorisations

5 Which of the following shows a correct prime factorisation in expanded form?

a $175 = 35 \times 5$

b $20 = 2 \times 2 \times 5$

c $286 = 2 \times 7 \times 11$

d $27 = 3^3$

6 Which of the following shows a correct prime factorisation in index form?

a $36 = 2 \times 2 \times 2 \times 3$

b $770 = 3^2 \times 55$

c $200 = 2^3 \times 5^2$

d $21 = 3^2 \times 7$

7 Write the following values as a product of prime factors in expanded form:

a 36

b 42

c 54

d 48

e 72

f 96

g 132

h 148

8 Write the following values as a product of prime factors in index form:

a 90

b 200

c 180

d 225

e 208

f 400

g 560

h 504

9 For the following values:

i Write as a product of prime factors in expanded form.

ii Using your answer from part (i), list all the factors.

a 12

b 35

c 20

d 40

e 52

f 54

g 66

h 275

10 List all the factors of 100.

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